

Harry Clemence

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Birmingham, UK

WORK EXPERIENCE

Blenheim Palace

Data Science Intern

July 2024 – September 2024

Woodstock, Oxfordshire

- Designed and implemented a predictive AI staff allocation model in Python, to schedule staff for customer-facing roles based on demand forecasts.
 - The model was integrated into Blenheim's industry leading Smart Visitor Management system, built using Power BI, to enable staff budgeting based on forecasts from the visitor prediction model.
 - Worked in collaboration with academic partners from Oxford Brookes University, and business stakeholders, including the CFO and Managing Director of Visitor Attractions, by communicating ideas and strategies through data visualisation skills.
 - Tech stack included AI and Machine Learning Python libraries such as Sci-Kit Learn, Pandas, Tensorflow, and SciPy, and data visualisation libraries including Matplotlib and Seaborn.
- Refined the AI visitor prediction model by improving feature extraction, accounting for patterns and interactions between events at Blenheim Palace and their durations.
 - Improved accuracy by up to 20%, with performance varying based on the prediction lead time.

International Tree Foundation

Software Consultant

September 2024

Oxford

- Implemented an automated workflow for grant managers and clients.
- Introduced a process for automatically generating contracts and securely sending them for signing, reducing administrative time and enhancing security.

UE Coffee Roasters

Barista

November 2023 – June 2024

Witney, Oxfordshire

PROJECTS

Frontend Re-build for Consolidate.org.uk

- Re-built a responsive landing page, about page and database pages according to Figma prototypes. Used a stack of Svelte, TypeScript, Tailwind and HTML.

Digital Twin Full-stack Application for Blenheim Palace

- Built a JavaScript frontend to render a digital model of the palace with a user interface and a Node.js REST API to connect the model to real-time data from IoT sensors and external data sources stored on a PostgreSQL database.

EDUCATION

University of Birmingham

2:1 BSc Computer Science

2020 – 2023

- Final Year Project involved the design and implementation of a Drum Machine React application with a Node.js REST API that ran a Generative neural network VAE to apply a humanised groove to a drum track.

Magdalen College School

2015 – 2020

- A-Levels: Maths (A*), Physics (A), Economics (A), Further Maths (B)